

TELESCOPING KELLER MAPS OF AFFINE THREE-SPACE: DEGREES AND EXPLICIT SYMMETRIC MONODROMY

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ABSTRACT. We give a telescoping construction of polynomial self-maps of affine three-space with Jacobian determinant -2 . Its exact polynomiality criterion consists of three scalar identities at $a = h(0) \neq 0$, and a member defined by a polynomial E of degree $n \geq 1$ has geometric degree exactly $n + 2$. Shifted Legendre polynomials supply members for every $n \geq 1$. An explicit degree-five member has primitive generic-fiber equation

$$\nu^5 - 5\nu^3 + 4\nu^2 - bc\nu + 2ac^2 = 0$$

and geometric monodromy S_5 ; hence its generic algebraic inverse is not solvable by radicals. We also record an S_4 member and the resulting degree spectrum $\mathbb{N} \setminus \{2\}$ in every dimension at least three. The exclusion of degree two is the already-known quadratic corollary of the classical Galois case of the Jacobian problem, not a new result here.

1. SCOPE AND CONVENTIONS

A *Keller map* is a polynomial map with nonzero constant Jacobian determinant, terminology originating with Keller's formulation of the Jacobian problem [4]. Its *geometric degree* is the degree of the induced extension of rational function fields, equivalently the cardinality of a general geometric fiber. We work over $\mathbb{C}$, although the displayed identities are defined over $\mathbb{Q}$.

The claims below are deliberately limited to the construction, its degree, and the two explicitly certified monodromy groups. Contemporaneous public work also reported constructions in every generic fiber degree at least three [3]; we record that overlap neutrally and make no claim here about exact priority or novelty. The accompanying script performs exact symbolic and finite-field checks.

2. THE TELESCOPING CONSTRUCTION

Put

$$s = x^2z, \quad t = xy, \quad u = 1 + t.$$

For $E(\nu) \in \mathbb{C}[\nu]$, define primitives with zero constant term

$$A(\nu) = \int_0^\nu \xi E(\xi) d\xi, \quad C(\nu) = 2 \int_0^\nu E(\xi) d\xi.$$

Given $h(t) \in \mathbb{C}[t]$, set

$$g = h(t) - s, \quad \nu = ug, \quad S = A(ug) + ug^2, \quad T = C(ug) + 2g.$$

Consider the initially rational expression

$$(1) \quad F_{E,h}(x, y, z) = \left(\frac{S(x^2z, xy)}{x^2g(x^2z, xy)^2}, \frac{T(x^2z, xy)}{xg(x^2z, xy)}, xg(x^2z, xy) \right).$$

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Theorem 2.1 (Telescoping family). *Let E have degree $n \geq 1$ and put $a = h(0)$. The expression (1) is a polynomial map if and only if $a \neq 0$ and*

$$(2) \quad A(a) = -a^2,$$

$$(3) \quad C(a) = -2a,$$

$$(4) \quad 1 + E(a) + h'(0) \frac{E(a) + 2}{a} = 0.$$

When these conditions hold,

$$\det \text{Jac}(F_{E,h}) = -2, \quad \deg_{\text{geom}}(F_{E,h}) = n + 2.$$

Proof. First suppose that $a \neq 0$. Write

$$H_1 = \frac{S}{g^2}, \quad H_2 = \frac{T}{g}.$$

Because A has no terms below degree two and C has no constant term,

$$H_1 = \sum_{k \geq 2} A_k u^k g^{k-2} + u, \quad H_2 = \sum_{k \geq 1} C_k u^k g^{k-1} + 2,$$

so $H_1, H_2 \in \mathbb{C}[s, t]$. A monomial $s^i t^j$ in $H_m(x^2 z, xy)/x^m$ produces $x^{2i+j-m} z^i y^j$. Thus polynomiality is equivalent to the absence of weight less than 2 in H_1 and weight less than 1 in H_2 . The only possible offenders are the constant terms of H_1, H_2 and the coefficient of t in H_1 . At $(s, t) = (0, 0)$ these vanish respectively exactly when (2), (3), and, after differentiating $A((1+t)h(t))/h(t)^2 + 1 + t$, (4) hold.

Conversely, if (1) is polynomial, the forbidden weight-zero terms of H_1 and H_2 must vanish. If $a = 0$, use $g = h(t) - s$ and evaluate along $t = 0$. The polynomial quotients are $S/g^2 = A(g)/g^2 + 1$ and $T/g = C(g)/g + 2$; polynomiality after division by x^2 and x forces their constant terms at $s = 0$ to vanish. In terms of $e_0 = E(0)$ these are $e_0/2 + 1 = 0$ and $2e_0 + 2 = 0$, which demand $e_0 = -2$ and $e_0 = -1$, an impossibility. Hence $a \neq 0$, and the preceding weight argument gives all three conditions. This also explains the degree-zero case: when $E = e$ is constant and $a \neq 0$, (2) gives $e = -2$ while (3) gives $e = -1$. The constraints are inconsistent; they do not “force $a = 0$.”

For the Jacobian, introduce $w = 1/u$. In coordinates (ν, w) ,

$$S = A(\nu) + \nu^2 w, \quad T = C(\nu) + 2\nu w.$$

Using $A' = \nu E$ and $C' = 2E$ gives

$$\frac{\partial(S, T)}{\partial(\nu, w)} = 2\nu^2 w, \quad \frac{\partial(\nu, w)}{\partial(s, t)} = u^{-1}, \quad \frac{\partial(S, T)}{\partial(s, t)} = 2g^2.$$

On $x \neq 0$, compare the volume forms in coordinates (x, s, t) and use the third target coordinate xg ; the $x dg$ term disappears when wedged with $ds \wedge dt$. One obtains

$$F^*(dw_1 \wedge dw_2 \wedge dw_3) = -2 dx \wedge dy \wedge dz.$$

As F is polynomial, the identity holds everywhere.

It remains to determine the degree. The change

$$(s, t) \mapsto (\nu, w) = ((1+t)(h(t) - s), (1+t)^{-1})$$

is birational, with $t = w^{-1} - 1$ and $s = h(w^{-1} - 1) - \nu w$. Eliminating w from $S = \alpha$ and $T = \beta$ gives

$$(5) \quad A(\nu) - \frac{1}{2}\nu C(\nu) + \frac{\beta}{2}\nu = \alpha.$$

If e_n is the leading coefficient of E , then the coefficient of ν^{n+2} in $2A - \nu C$ is

$$2e_n \left(\frac{1}{n+2} - \frac{1}{n+1} \right) = -\frac{2e_n}{(n+1)(n+2)} \neq 0.$$

Hence (5) has degree exactly $n+2$ and has $n+2$ simple roots generically. This reconstruction is made on the dense open set where the birational chart and its denominators are defined: in particular $\nu, u, g, c \neq 0$. A root $\nu = 0$ would force $\alpha = 0$ in (5), so it creates no extraneous point over a general target. Each remaining root reconstructs a unique (s, t) , and one then uniquely recovers $x = c/g$, $y = t/x$, and $z = s/x^2$. Therefore the geometric degree is exactly $n+2$. $\square$

3. EXISTENCE IN EVERY DEGREE AT LEAST THREE

Let L_n be the Legendre polynomial normalized by $L_n(1) = 1$.

Lemma 3.1 (Shifted Legendre moments). *For $n \geq 1$,*

$$\int_0^1 L_n(2\nu - 1) d\nu = 0,$$

and for $n \geq 2$,

$$\int_0^1 \nu L_n(2\nu - 1) d\nu = 0.$$

Moreover $L_n(1) = 1$.

Proof. After $r = 2\nu - 1$, the first integral is one half of $\int_{-1}^1 L_n(r) dr$, which vanishes by orthogonality to the constant polynomial for $n \geq 1$. The second is one quarter of $\int_{-1}^1 (r+1)L_n(r) dr$, which vanishes for $n \geq 2$ by orthogonality to every polynomial of degree at most one. The endpoint normalization is part of the standard definition (and follows immediately from Rodrigues' formula). $\square$

Theorem 3.2 (Universal existence). *For every $n \geq 1$ there are E of degree n and h satisfying Theorem 2.1. Consequently every integer $d \geq 3$ is the geometric degree of a Keller self-map of $\mathbb{A}_{\mathbb{C}}^3$ with determinant -2 .*

Proof. For $n = 1$, take

$$E(\nu) = 2 - 6\nu, \quad h(t) = 1 - \frac{3}{2}t.$$

Then $A(1) = -1$, $C(1) = -2$, $E(1) = -4$, and $1 - 4 + (-3/2)(-2) = 0$.

For $n \geq 2$, take

$$(6) \quad E(\nu) = 2 - 6\nu + L_n(2\nu - 1), \quad h(t) = 1 - 2t.$$

The two identities of Lemma 3.1 give

$$C(1) = 2 \int_0^1 E = -2, \quad A(1) = \int_0^1 \nu E = -1.$$

Also

$$E(1) = 2 - 6 + L_n(1) = -3;$$

hence the third condition is $1 - 3 + (-2)(-1) = 0$. In all cases $a = h(0) = 1 \neq 0$, so Theorem 2.1 applies and gives degree $n+2$. $\square$

4. AN EXPLICIT QUINTIC WITH GEOMETRIC MONODROMY S_5

Take $E = 4 - 15\nu + 10\nu^3$ and $h = 1$. Thus

$$A(\nu) = 2\nu^2 - 5\nu^3 + 2\nu^5, \quad C(\nu) = 8\nu - 15\nu^2 + 5\nu^4, \quad g = 1 - x^2z, \quad \nu = (1 + xy)g.$$

The map is compactly

$$(7) \quad F_5 = \left(\frac{A(\nu) + (1 + xy)g^2}{x^2g^2}, \frac{C(\nu) + 2g}{xg}, xg \right).$$

Theorem 2.1 proves polynomiality and gives $\det \text{Jac}(F_5) = -2$ and $\deg_{\text{geom}} F_5 = 5$.

Let (a, b, c) denote target coordinates. Since $c = xg$, the fiber equations imply

$$ac^2 = A(\nu) + \nu g, \quad bc = C(\nu) + 2g.$$

Eliminating g gives the primitive equation

$$(8) \quad P_{a,b,c}(\nu) = \nu^5 - 5\nu^3 + 4\nu^2 - bc\nu + 2ac^2 = 0.$$

Conversely, away from the proper loci where a denominator vanishes, a root reconstructs the source point by

$$(9) \quad g = \frac{bc - C(\nu)}{2}, \quad x = \frac{c}{g}, \quad u = \frac{\nu}{g}, \quad y = \frac{u - 1}{x}, \quad z = \frac{1 - g}{x^2}.$$

These are exact rational identities modulo (8).

Theorem 4.1. *The arithmetic and geometric monodromy groups of F_5 are S_5 . Consequently the generic algebraic inverse is not solvable by radicals over $\mathbb{C}(a, b, c)$.*

Proof. At the targets

$$(2/3, -1/5, 3/4), \quad (1/2, 1/3, -2/7), \quad (-3/5, 4/9, 1/6),$$

the primitive integral specializations of (8) are

$$\begin{aligned} Q_1 &= 20\nu^5 - 100\nu^3 + 80\nu^2 + 3\nu + 15, \\ Q_2 &= 147\nu^5 - 735\nu^3 + 588\nu^2 + 14\nu + 12, \\ Q_3 &= 270\nu^5 - 1350\nu^3 + 1080\nu^2 - 20\nu - 9. \end{aligned}$$

Exact finite-field factorizations include

$$\begin{aligned} Q_1 \bmod 7 &= 6(\nu^5 + 2\nu^3 - 3\nu^2 - 3\nu - 1), & \text{irreducible,} \\ Q_1 \bmod 103 &= 20(\nu + 11)(\nu + 18)(\nu + 21)(\nu^2 - 50\nu + 40). \end{aligned}$$

Both primes are good: the specialized discriminant is nonzero modulo each, so the reductions are squarefree and the primes are unramified in the splitting field of Q_1 . Their Frobenius cycle types show that $\text{Gal}(Q_1/\mathbb{Q})$ contains a 5-cycle and a transposition. Irreducibility modulo 7 also makes Q_1 irreducible over $\mathbb{Q}$, so this group is transitive; transitivity in prime degree implies primitivity, and a primitive subgroup of S_5 containing a transposition is S_5 . Since this rational target lies off the discriminant and denominator loci, the specialization theorem embeds its Galois group, up to conjugacy, into the generic arithmetic monodromy group. The latter is a subgroup of S_5 containing S_5 , hence is S_5 .

For geometric monodromy, (8) is irreducible over $\mathbb{C}(a, b, c)$. Indeed, over $K = \mathbb{C}(b, c)$ it is monic and primitive in $K[a][\nu]$, and it has degree one in a with nonzero coefficient $2c^2 \in K^*$. In a factorization, one factor has a -degree zero; comparison of the coefficient of a makes that factor divide a unit of K , so it is a unit. Hence the geometric group is transitive and normal in the arithmetic S_5 , leaving A_5 or S_5 .

The three exact specialized discriminants are

$$\begin{aligned}\Delta_1 &= 797038565664000, \\ \Delta_2 &= -244036829099277031104, \\ \Delta_3 &= 17125713993062891250000.\end{aligned}$$

Neither $\Delta_1\Delta_2$ nor $\Delta_1\Delta_3$ is a rational square. If the geometric group lay in A_5 , the generic discriminant over $\mathbb{Q}(a, b, c)$ would be a constant times a rational square, forcing every product of two good rational specializations to be a rational square. This contradiction gives geometric group S_5 .

Finally, a radical expression for the generic algebraic inverse would put the extension generated through (9) inside a radical tower. Its Galois closure would then have solvable group, contrary to S_5 . $\square$

Remark 4.2. The last conclusion concerns the generic algebraic inverse. It does not deny local holomorphic inverses: since $\det \text{Jac}(F_5) = -2$, the holomorphic inverse function theorem supplies a holomorphic inverse germ at every point.

5. THE DEGREE-FOUR MEMBER

For $E = 3 - 12\nu + 6\nu^2$ and $h = 1 - 2t$, Theorem 2.1 gives a degree-four Keller map F_4 . Its primitive generic-fiber equation is

$$(10) \quad \nu^4 - 4\nu^3 + 3\nu^2 - bc\nu + 2ac^2 = 0.$$

Proposition 5.1. *The arithmetic and geometric monodromy groups of F_4 are S_4 .*

Proof. At $(a, b, c) = (2/3, -1/5, 3/4)$ the primitive polynomial is

$$R = 20\nu^4 - 80\nu^3 + 60\nu^2 + 3\nu + 15.$$

Its exact factorizations have types (4) modulo 11, (1, 3) modulo 7, and (1, 1, 2) modulo 19; the discriminant is nonzero modulo each prime, so these are good, unramified reductions. Irreducibility modulo 11 proves that R is irreducible over $\mathbb{Q}$, hence its Galois group is transitive and contains a 4-cycle. The reduction modulo 7 supplies a 3-cycle. The group order is therefore divisible by 12; among transitive subgroups of S_4 , a subgroup of order divisible by 12 is A_4 or S_4 , and the 4-cycle excludes A_4 . Thus $\text{Gal}(R/\mathbb{Q}) = S_4$ (the reduction modulo 19 additionally supplies a transposition). The target is off the discriminant and denominator loci, so specialization embeds this group, up to conjugacy, into the generic arithmetic monodromy group; consequently that generic group is also S_4 .

The generic quartic (10) is irreducible over $\mathbb{C}(a, b, c)$: over $K = \mathbb{C}(b, c)$ it is monic and primitive in $K[a][\nu]$, linear in a , and its a -coefficient $2c^2$ is a unit of K . In any factorization one factor has a -degree zero, and comparison of the a -coefficient makes it divide that unit. Thus the geometric group is transitive; it is also normal in the arithmetic S_4 , so it is V_4 , A_4 , or S_4 . In the V_4 case the degree-four extension would itself be Galois (the transitive V_4 action is regular). Campbell's Galois case over $\mathbb{C}$ says that a polynomial map with nonzero constant Jacobian and Galois induced function-field extension has a polynomial inverse [1, 2], contradicting degree four. To exclude A_4 , use the three specializations above; their discriminants are

$$-54873514800, \quad -466029671480352, \quad 4549902124608000.$$

The product of the first and third is negative and hence not a rational square. The same constant-times-square specialization argument used for S_5 excludes A_4 . Therefore the geometric group is S_4 . $\square$

6. DEGREE SPECTRUM

We take $\mathbb{N} = \{1, 2, 3, \dots\}$.

Corollary 6.1. *For every $N \geq 3$, the geometric-degree spectrum of Keller self-maps of $\mathbb{A}_{\mathbb{C}}^N$ is*

$$\mathbb{N} \setminus \{2\} = \{1, 3, 4, 5, \dots\}.$$

Proof. Degree one is realized by automorphisms. Every degree $d \geq 3$ occurs on $\mathbb{A}^3$ by Theorem 3.2, and adjoining $N - 3$ identity coordinates preserves the degree and the Keller property.

For completeness, degree two cannot occur. A separable extension of degree two is Galois, and the classical Galois case over $\mathbb{C}$ [1, 2] then makes the Keller map invertible, hence of degree one. This degree-two exclusion is an already-known corollary of the classical theorem and is not claimed as new. $\square$

7. REPRODUCIBILITY

The file `verify.py` uses Python 3 and SymPy exact arithmetic. It verifies the shifted-Legendre constraints for a range of indices, the polynomiality and Jacobian identities for representative family members, the quintic and quartic primitive equations and rational reconstruction, all stated finite-field certificates, and all specialized discriminants. A successful run ends with **ALL CHECKS PASSED**. No numerical root counts are used.

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